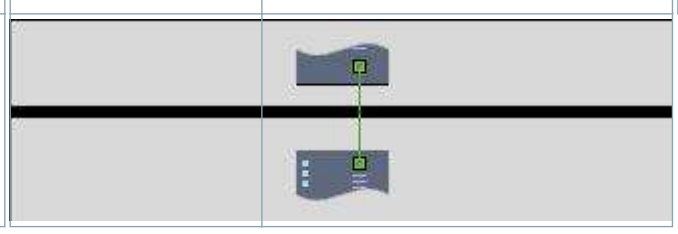
PLC_2 [CPU 1211C DC/DC/DC]

PLC_2					
General\Project information					
Name	PLC_2	**Author**	praktykant	**Comment**	
Slot	1	**Rack**	0		
General\Catalog information					
Short designation	CPU 1211C DC/DC/DC	**Description**	Work memory 75 KB; 24VDC power supply with DI6 x 24VDC SINK/SOURCE, DQ4 x 24VDC and AI2 on board; 3 high-speed counters (expandable with digital signal board) and 4 pulse outputs on board; signal board expands on-board I/O; up to 3 communication modules for serial communication; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA	**Article number**	6ES7 211-1AE40-0XB0
Firmware version	V4.6		False		
General\Identification & Maintenance					
Plant designation		**Location identifier**		**Installation date**	2025-07-14 09:19:44.558
Additional information					
General\Checksums					
Text lists	FA 70 E8 75 1D 5A 8E 29	**Software**	EE 93 4A 0E 3B 14 21 07		
PROFINET interface [X1]\General					
Name	PROFINET interface_1	**Author**	praktykant	**Comment**	
PROFINET interface [X1]\General\Project information					
Name	DI 6/DQ 4_1	**Comment**		**Name**	AI 2_1
Comment					
PROFINET interface [X1]\Ethernet addresses\Interface networked with					
Subnet:	PN/IE_1				
PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)					
IP configuration	Set IP address in the project	**IP address:**	10.10.1.50	**Subnet mask:**	255.255.255.0
Use router	False				
PROFINET interface [X1]\Ethernet addresses\PROFINET					
PROFINET device name is set directly at the device	False	**Generate PROFINET device name automatically**	True	**PROFINET device name:**	plc_2
Converted name:	plcxb2d1ad	**Device number:**	0		
PROFINET interface [X1]\Time synchronization					
Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses	**Server 1**	0.0.0.0
Server 2	0.0.0.0	**Server 3**	0.0.0.0	**Server 4**	0.0.0.0
Update interval	10sec			**CPU synchronizes the modules of the device.**	No synchronization
PROFINET interface [X1]\Digital inputs\Channel0					
Channel address	I0.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel0					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49152	**Event name:**	0
Hardware interrupt:	0	**Rising edge0**	Rising edge0		
PROFINET interface [X1]\Digital inputs\Channel0					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49280	**Event name:**	0
Hardware interrupt:	0	**Falling edge0**	Falling edge0		
PROFINET interface [X1]\Digital inputs\Channel1					
Channel address	I0.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel1					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49153	**Event name:**	0
Hardware interrupt:	0	**Rising edge1**	Rising edge1		
PROFINET interface [X1]\Digital inputs\Channel1					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49281	**Event name:**	0
Hardware interrupt:	0	**Falling edge1**	Falling edge1		
PROFINET interface [X1]\Digital inputs\Channel2					
Channel address	I0.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel2					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49154	**Event name:**	0
Hardware interrupt:	0	**Rising edge2**	Rising edge2		
PROFINET interface [X1]\Digital inputs\Channel2					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49282	**Event name:**	0
Hardware interrupt:	0	**Falling edge2**	Falling edge2		
PROFINET interface [X1]\Digital inputs\Channel3					
Channel address	I0.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel3					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49155	**Event name:**	0
Hardware interrupt:	0	**Rising edge3**	Rising edge3		

PROFINET interface [X1]\Digital inputs\Channel3					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49283	**Event name:**	0
Hardware interrupt:	0	**Falling edge3**	Falling edge3		
PROFINET interface [X1]\Digital inputs\Channel4					
Channel address	I0.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel4					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49156	**Event name:**	0
Hardware interrupt:	0	**Rising edge4**	Rising edge4		
PROFINET interface [X1]\Digital inputs\Channel4					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49284	**Event name:**	0
Hardware interrupt:	0	**Falling edge4**	Falling edge4		
PROFINET interface [X1]\Digital inputs\Channel5					
Channel address	I0.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel5					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49157	**Event name:**	0
Hardware interrupt:	0	**Rising edge5**	Rising edge5		
PROFINET interface [X1]\Digital inputs\Channel5					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49285	**Event name:**	0
Hardware interrupt:	0	**Falling edge5**	Falling edge5		
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				
PROFINET interface [X1]\Digital outputs\Channel0					
Channel address	Q0.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel1					
Channel address	Q0.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel2					
Channel address	Q0.2	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel3					
Channel address	Q0.3	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Operating mode					
IO controller	True	**IO system**		**Device number**	0
IO device	False				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	0.0	**End address**	0.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	64	**End address**	67	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Output addresses					
Start address	0.0	**End address**	0.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\Advanced options\Interface options					
Support device replacement without exchangeable medium	True	**Permit overwriting of device names of all assigned IO devices**	False	**Use IEC V2.2 LLDP mode**	False
Keep-Alive connection monitoring:	30s				
PROFINET interface [X1]\Advanced options\Real time settings\IO communication					
Send clock:	1.000ms				
PROFINET interface [X1]\Advanced options\Real time settings\Real time options					
Calculated bandwidth for cyclic IO data:	0.000ms	**Calculated bandwidth for cyclic IO data:**	0.000%		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General					
Name	Port_1	**Author**	praktykant	**Comment**	
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:					
Local port:	PLC_2\PROFINET interface_1 [X1]\Port_1 [X1 P1]	**Medium:**	Copper	**Cable name:**	---
					

PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:					
	Monitoring of partner port is not possible	**Partner port:**	Switch_1\SCALANCE interface_1 [X1]\Port_3 [X1 P3]	**Medium:**	Copper
Cable length:					
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate					
Activate this port for use	True				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection					
Transmission rate / duplex:	Automatic	**Monitor**	False	**Enable autonegotiation**	True
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries					
End of detection of accessible devices	False	**End of topology discovery**	False	**End of the sync domain**	False
PROFINET interface [X1]\Web server access					
Enable Web server for the IP address of this interface	False	**The Web server must also be activated in the properties of the PLC.**			
High speed counters (HSC)\HSC1\General\Enable					
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
High speed counters (HSC)\HSC1\General\Project information					
Name	HSC_1	**Comment**		**Name**	HSC_2
Comment		**Name**	HSC_3	**Comment**	
Name	HSC_4	**Comment**		**Name**	HSC_5
Comment		**Name**	HSC_6	**Comment**	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses					
Start address	1000.0	**End address**	1003.7	**Start address**	1004.0
End address	1007.7	**Organization block**	0	**Start address**	1008.0
End address	1011.7	**Organization block**	0	**Process image**	0
Start address	1012.0	**End address**	1015.7	**Organization block**	0
Process image	0	**Start address**	1016.0	**End address**	1019.7
Organization block	0	**Process image**	0	**Start address**	1020.0
End address	1023.7	**Organization block**	0	**Process image**	0
Organization block	0	**Process image**	0	**Process image**	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable					
Enable this pulse generator	0	**Enable this pulse generator**	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information					
Name	Pulse_1	**Comment**		**Name**	Pulse_2
Comment					
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses					
Start address	1000.0	**End address**	1001.7	**Start address**	1002.0
End address	1003.7	**Organization block**	0	**Organization block**	0
Process image	0	**Process image**	0		
Startup					
Startup after POWER ON	Warm restart - mode before POWER OFF	**Comparison preset to actual configuration**	Startup CPU even if mismatch	**Configuration time**	60000ms
OBs should be interruptible	1				
Cycle					
Cycle monitoring time [ms]	150ms			**Enable minimum cycle time for cyclic OBs**	0
Minimum cycle time	1ms				
Communication load					
Cycle load due to communication [%]	20%				
System and clock memory\System memory bits					
Enable the use of system memory byte	1	**Address of system memory byte (MBx)**	1	**First cycle**	%M1.0 (FirstScan)
Diagnostic status changed	%M1.1 (DiagStatusUpdate)	**Always 1 (high)**	%M1.2 (AlwaysTRUE)	**Always 0 (low)**	%M1.3 (AlwaysFALSE)
System and clock memory\Clock memory bits					
Enable the use of clock memory byte	0	**Address of clock memory byte (MBx)**	0	**10 Hz clock**	
5 Hz clock		**2.5 Hz clock**		**2 Hz clock**	
1.25 Hz clock		**1 Hz clock**		**0.625 Hz clock**	
0.5 Hz clock					
Web server\General					
Activate Web server on all modules of this device	False	**Permit access only with HTTPS**	True		
Web server\Automatic update					
Enable automatic update	True	**Update interval**	0s		

Web server\User management

User name	User rights
Everybody	

Web server\User-defined web pages

Application name	HTML source path	Default HTML page	Files with dynamic content	Web DB number	Fragment DB number
		index.htm	.htm;.html	333	334

Web server\Overview of interfaces

Device	Interface	Enabled web server access
PLC_2	PROFINET interface_1	False

User interface languages	
Assign project language	**User interface languages**
English (United States)	German
English (United States)	English
English (United States)	French
English (United States)	Spanish
English (United States)	Italian
English (United States)	Chinese (simplified)

Time of day\Local time					
Time zone	(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna				
Time of day\Daylight saving time					
Activate daylight saving time	1	**Difference between standard and daylight saving time**	60min		
Time of day\Daylight saving time\Start of daylight saving time					
First week of the month:	Last		Sunday	**in**	March
at	1:00 a.m.				
Time of day\Daylight saving time\Start of standard time					
	Last		Sunday	**in**	October
at	2:00 a.m.				
Protection & Security					
Level of protection	No protection				
Protection & Security\Connection mechanisms					
Permit access with PUT/GET communication from remote partner	True				
Protection & Security\Security event					
Summarize diagnostics in case of high message volume	True	**Length of an interval**	20	**Unit**	seconds
Protection & Security\External load memory					
Disable copying from internal load memory to external load memory	False				
Advanced configuration\DnsParameterConfigurationMenu					
The Tree-Node DnsConfigurationMenu was not filled by some ACF					
Advanced configuration\Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).					
Activate SNMP	False				
Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				

Connection resources

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_2 [CPU 1211C DC/DC/DC] - Configured
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	2	0	2
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		32	34	66

Overview of addresses\Overview of addresses\Overview of addresses					
Inputs	True	**Outputs**	True	**Address gaps**	False
Slot	True				

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	0	DI 6/DQ 4_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	1 Bytes	-	0	1 1
O	0	0	DI 6/DQ 4_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	1 Bytes	-	0	1 1
I	64	67	AI 2_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 35

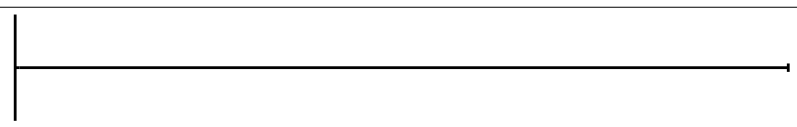
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

Main [OB1]

Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1:



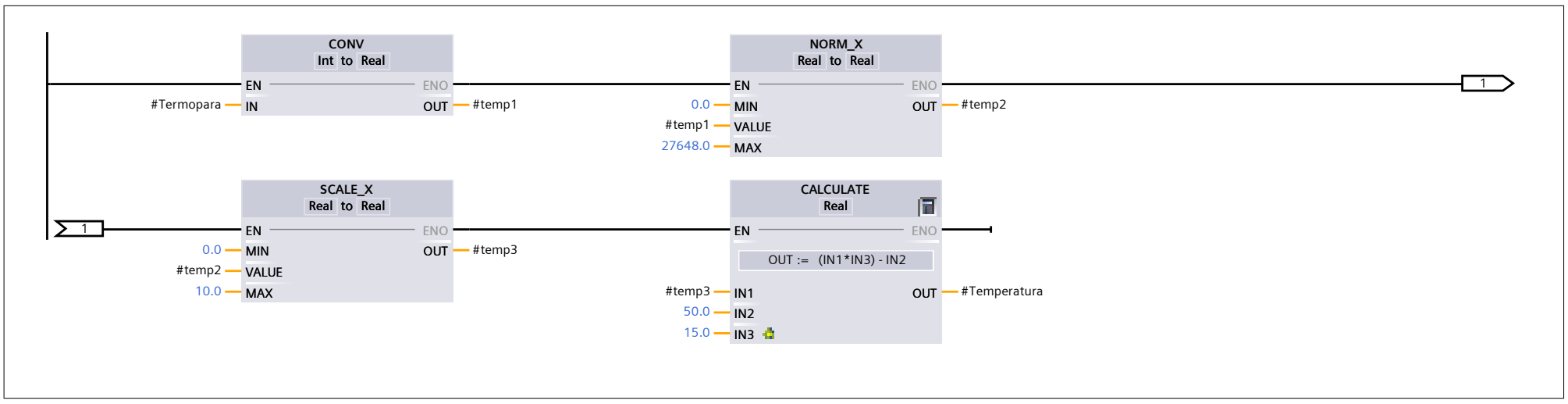
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

Skalowanie [FC1]

Skalowanie Properties							
General							
Name	Skalowanie	**Number**	1	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Termopara	Int		
Output			
Temperatura	Real		
InOut			
Temp			
temp1	Real		
temp2	Real		
temp3	Real		
Constant			
Return			
Skalowanie	Void		

Network 1:



PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

Temperatura_DB [DB1]

Temperatura_DB Properties							
General							
Name	Temperatura_DB	**Number**	1	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Offset	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Static										
temperatura	Real	0.0	0.0	True	True	True	True	False		
test	Real	4.0	0.0	True	True	True	True	False		
ON_OFF	Bool	8.0	false	True	True	True	True	False		
test-bool	Bool	8.1	false	True	True	True	True	False		
emergency	Bool	8.2	false	True	True	True	True	False		
automatic_manual	Bool	8.3	false	True	True	True	True	False		
ON_OFF_activate	Bool	8.4	false	True	True	True	True	False		
HEATER_ON	Bool	8.5	false	True	True	True	True	False		
test_1	Real	10.0	0.0	True	True	True	True	False		
setpoint	Real	14.0	0.0	True	True	True	True	False		
output	Real	18.0	0.0	True	True	True	True	False		
cyclicMode	Bool	22.0	false	True	True	True	True	False		
P	Real	24.0	0.0	True	True	True	True	False		
I	Real	28.0	0.0	True	True	True	True	False		
D	Real	32.0	0.0	True	True	True	True	False		
manual_value	Real	36.0	0.0	True	True	True	True	False		

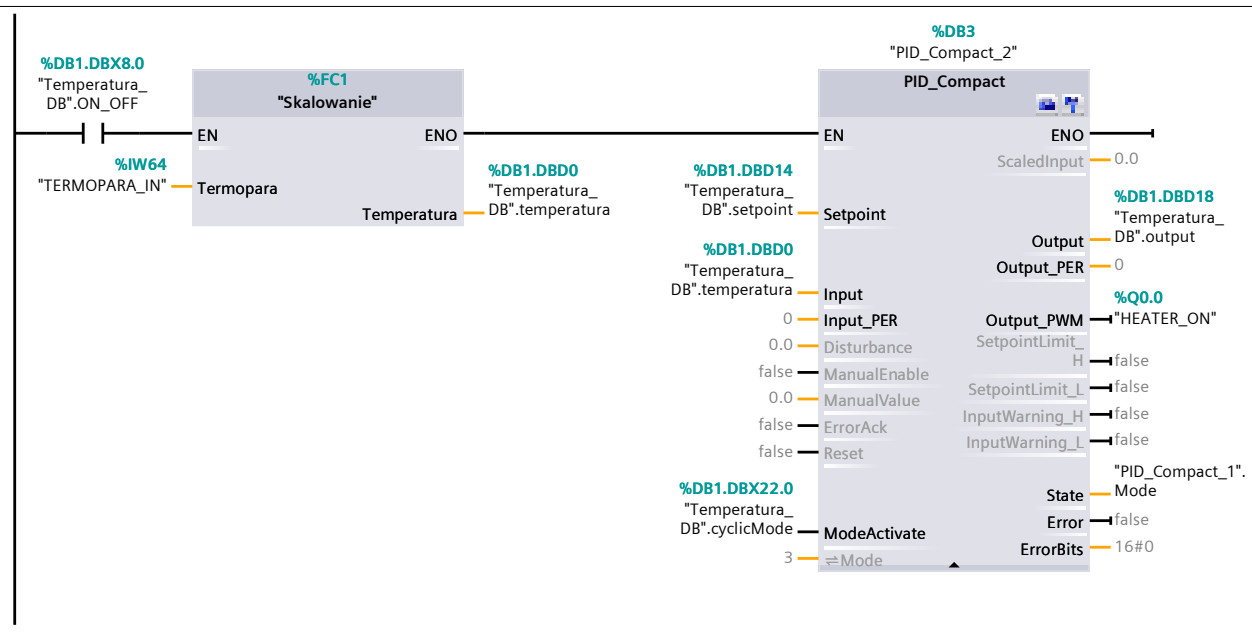
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

Main_alg [FC2]

Main_alg Properties							
General							
Name	Main_alg	**Number**	2	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
Main_alg	Void		

Network 1:



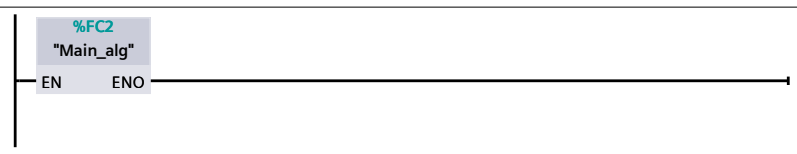
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

CyclicInterrupt [OB30]

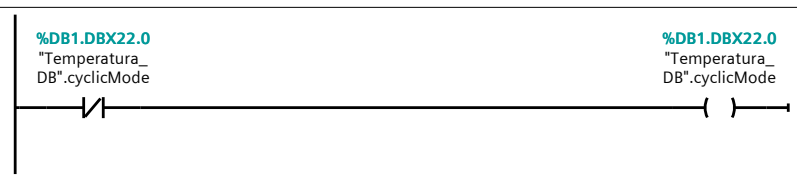
CyclicInterrupt Properties							
General							
Name	CyclicInterrupt	**Number**	30	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Event_Count	Int		Events discarded
Temp			
temp1	Bool		
Constant			

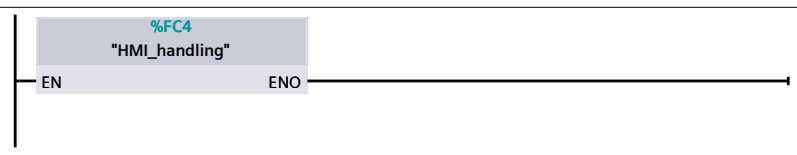
Network 1:



Network 2:



Network 3:



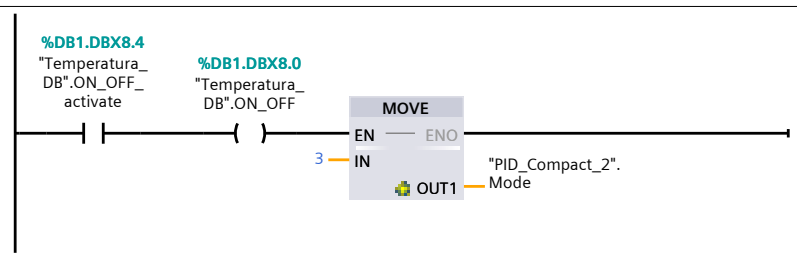
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

Test_ON_OFF [FC3]

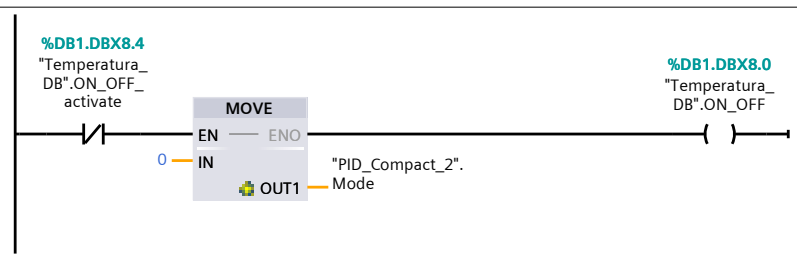
Test_ON_OFF Properties							
General							
Name	Test_ON_OFF	**Number**	3	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
Return			
Test_ON_OFF	Void		

Network 1:



Network 2:



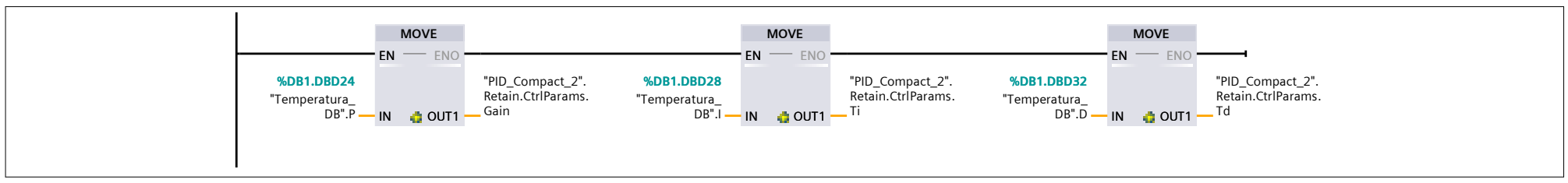
PLC_2 [CPU 1211C DC/DC/DC] / Program blocks

HMI_handling [FC4]

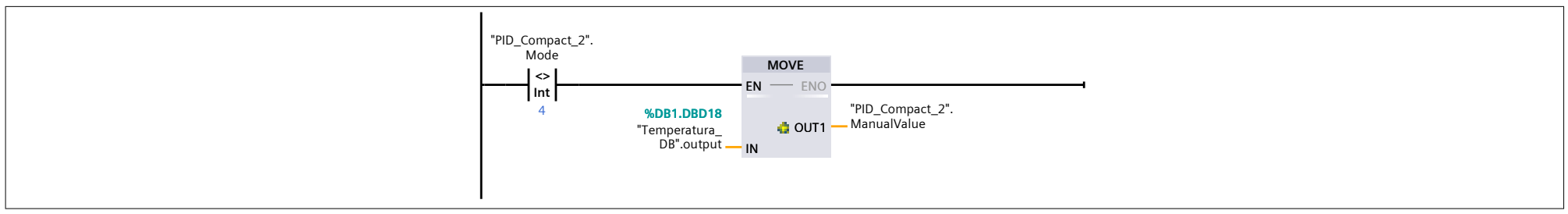
HMI_handling Properties							
General							
Name	HMI_handling	**Number**	4	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
▼ Temp			
temp1	Bool		
temp2	Bool		
temp3	Bool		
temp4	Bool		
Constant			
▼ Return			
HMI_handling	Void		

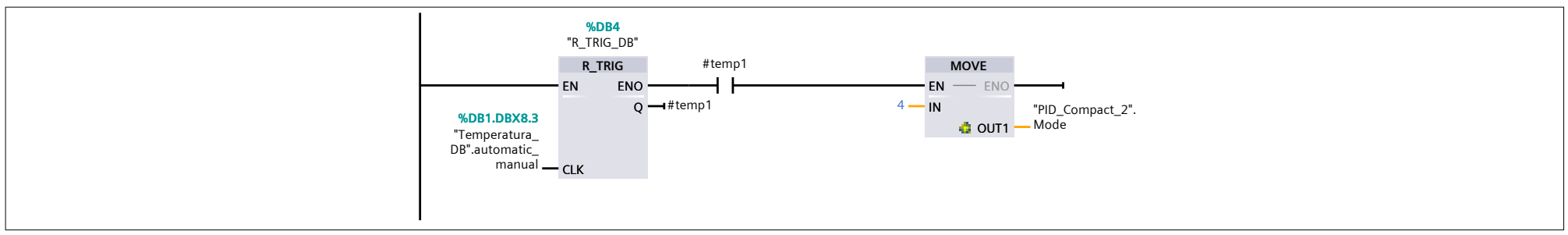
Network 1:



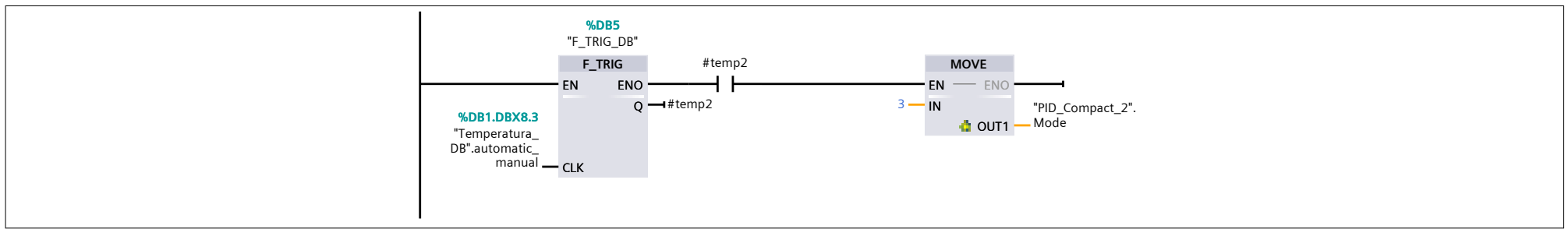
Network 2:



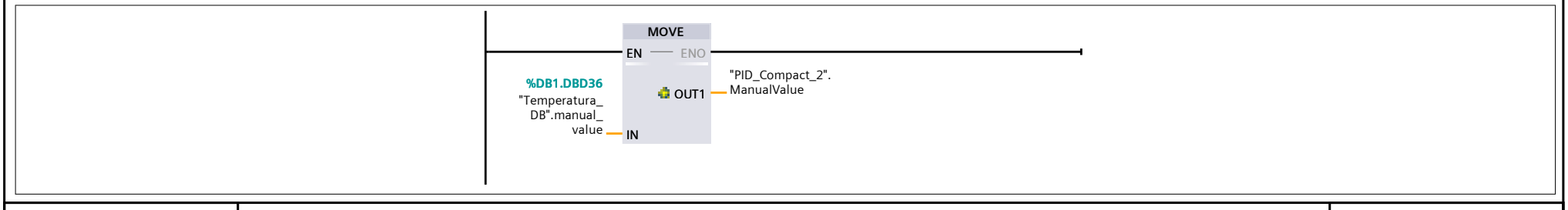
Network 3:



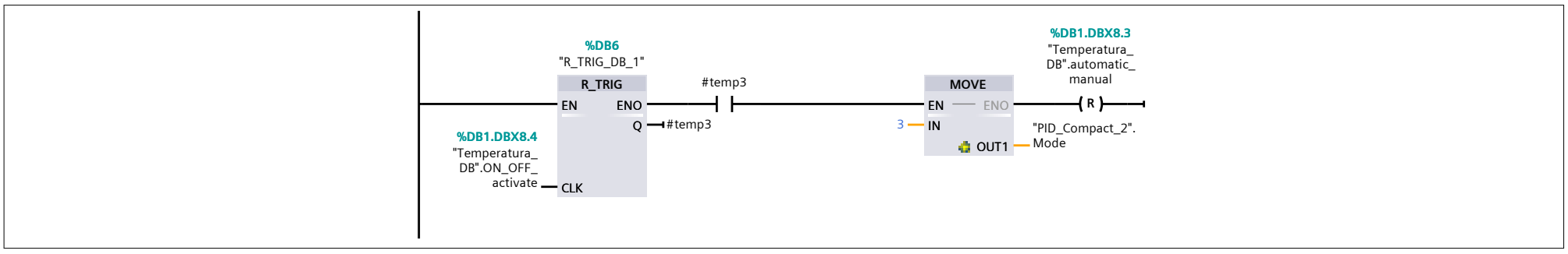
Network 4:



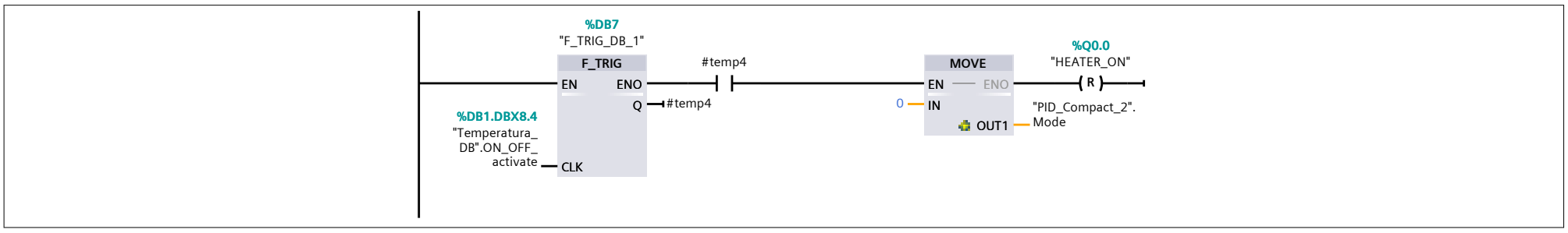
Network 5:



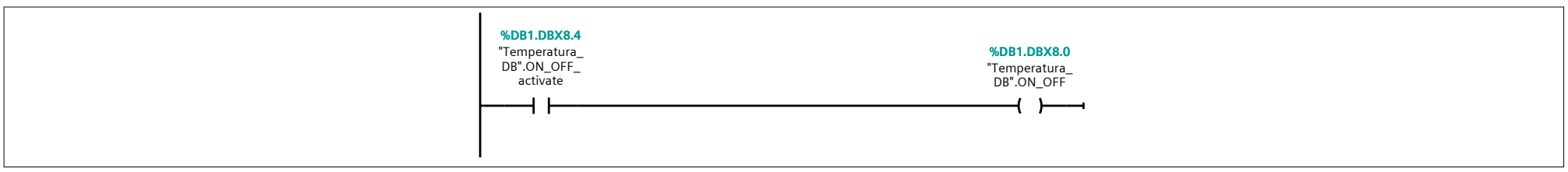
Network 6:



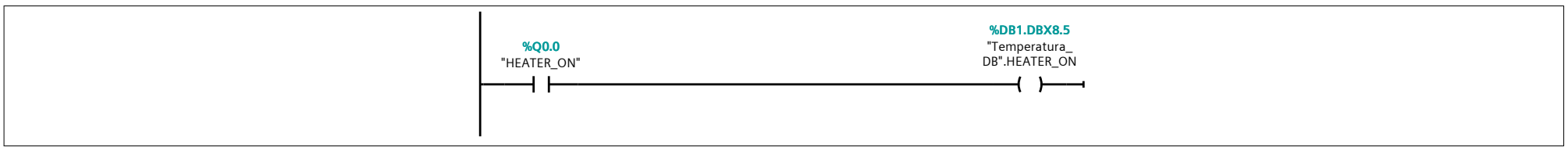
Network 7:



Network 8:



Network 9:



PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

PID_Compact [FB1130]

PID_Compact Properties							
General							
Name	PID_Compact	**Number**	1130	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title	Compact PID_Controller with self-tuning	**Author**	SIMATIC	**Comment**		**Family**	COMPPID
Version	2.3	**User-defined ID**	PID_Cmpt				

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Setpoint	Real	0.0	Non-retain	True	True	True	False		controller setpoint input
Input	Real	0.0	Non-retain	True	True	True	False		current value from process in REAL format
Input_PER	Int	0	Non-retain	True	True	True	False		current value from peripheral input
Disturbance	Real	0.0	Non-retain	True	True	True	False		disturbance intrusion
ManualEnable	Bool	false	Non-retain	True	True	True	False		activate manual value to overwrite output value
ManualValue	Real	0.0	Non-retain	True	True	True	False		manual value
ErrorAck	Bool	false	Non-retain	True	True	True	False		reset error message
Reset	Bool	false	Non-retain	True	True	True	False		reset the controller
ModeActivate	Bool	false	Non-retain	True	True	True	False		enable mode
Output									
ScaledInput	Real	0.0	Non-retain	True	False	True	False		current value after scaling
Output	Real	0.0	Non-retain	True	False	True	False		output value in REAL format
Output_PER	Int	0	Non-retain	True	False	True	False		analog output value
Output_PWM	Bool	false	Non-retain	True	False	True	False		pulse width modulated output value
SetpointLimit_H	Bool	false	Non-retain	True	False	True	False		setpoint reached upper limit
SetpointLimit_L	Bool	false	Non-retain	True	False	True	False		setpoint reached lower limit
InputWarning_H	Bool	false	Non-retain	True	False	True	False		current value reached upper warning level
InputWarning_L	Bool	false	Non-retain	True	False	True	False		current value reached lower warning level
State	Int	0	Non-retain	True	False	True	False		current mode of operation (0-Inactive, 1-SUT, 2-TIR, 3-Automatic, 4-Manual, 5-Substitute output)
Error	Bool	false	Non-retain	True	False	True	False		error flag
ErrorBits	DWord	16#0	Retain	True	False	True	False		error message
InOut									
Mode	Int	4	Retain	True	True	True	False		mode selection
Static									
InternalDiagnostic	DWord	0	Non-retain	False	False	False	False		internal diagnostic and version handling
InternalVersion	DWord	DW#16#02030003	Non-retain	True	False	True	False		version of controller
InternalRTVersion	DWord	0	Non-retain	False	False	False	False		version of runtime
IntegralResetMode	Int	4	Non-retain	True	True	True	True		0 smooth, 1 clear, 2 keep, 3 overwrite initial output, 4 like setpoint change
OverwriteInitialOutputValue	Real	0.0	Non-retain	True	True	True	False		initialisation of output value for override control
RunModeByStartup	Bool	true	Non-retain	True	True	True	True		activate Mode after CPU restart
LoadBackUp	Bool	false	Non-retain	True	True	True	False		restore last parameter set
SetSubstituteOutput	Bool	true	Non-retain	True	True	True	True		assignment of output value in State = 5 (FALSE - last valid value, TRUE - SubstituteOutput)
PhysicalUnit	Int	0	Non-retain	True	False	True	True		unit of measurement of the process value and setpoint
PhysicalQuantity	Int	0	Non-retain	True	False	True	True		physical quantity of the process value and setpoint
ActivateRecoverMode	Bool	true	Non-retain	True	True	True	True		FALSE - go to inactive by error, TRUE - activate error treatment
Warning	DWord	16#0	Retain	True	False	True	False		warning message
WarningInternal	DWord	16#0	Retain	True	False	True	False		warning message
Progress	Real	0.0	Non-retain	True	False	True	False		progress of current phase in percent
CurrentSetpoint	Real	0.0	Non-retain	True	False	True	False		current active setpoint value
CancelTuningLevel	Real	10.0	Non-retain	True	True	True	True		cancel level for setpoint change during tuning

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision				Comment
SubstituteOutput	Real	0.0	Non-retain	True	True	True	True					substitute output value in case of error
Config	PID_Compact-Config		Non-retain	True	True	True	True					configuration data set
InputPerOn	Bool	true	Non-retain	True	True	True	True					activate peripheral input
InvertControl	Bool	false	Non-retain	True	True	True	True					invert control direction
InputUpperLimit	Real	120.0	Non-retain	True	True	True	True					input (process value) upper limit
InputLowerLimit	Real	0.0	Non-retain	True	True	True	True					input (process value) lower limit
InputUpperWarning	Real	3.402822e+38	Non-retain	True	True	True	True					input (process value) upper level warning
InputLowerWarning	Real	-3.402822e+38	Non-retain	True	True	True	True					input (process value) lower level warning
OutputUpperLimit	Real	100.0	Non-retain	True	True	True	True					output value upper limit
OutputLowerLimit	Real	0.0	Non-retain	True	True	True	True					output value lower limit
SetpointUpperLimit	Real	3.402822e+38	Non-retain	True	True	True	True					setpoint upper limit value
SetpointLowerLimit	Real	-3.402822e+38	Non-retain	True	True	True	True					setpoint lower limit value
MinimumOnTime	Real	0.0	Non-retain	True	True	True	True					PWM minimum on time
MinimumOffTime	Real	0.0	Non-retain	True	True	True	True					PWM minimum off time
InputScaling	PID_Scaling		Non-retain	True	True	True	True					input scaling
UpperPointIn	Real	27648.0	Non-retain	True	True	True	True					high value (input range of scaling)
LowerPointIn	Real	0.0	Non-retain	True	True	True	True					low value (input range of scaling)
UpperPointOut	Real	100.0	Non-retain	True	True	True	True					high value (output range of scaling)
LowerPointOut	Real	0.0	Non-retain	True	True	True	True					low value (output range of scaling)
CycleTime	PID_CycleTime		Non-retain	True	True	True	True					data set for cycle time estimation
StartEstimation	Bool	true	Non-retain	True	True	True	False					start automatic estimation of call cycle time
EnEstimation	Bool	true	Non-retain	True	True	True	True					enable estimation of call cycle time
EnMonitoring	Bool	true	Non-retain	True	True	True	True					enable monitoring of call cycle time
Value	Real	0.1	Non-retain	True	True	True	True					call cycle time
CtrlParamsBackUp	PID_Compact-ControlParams		Non-retain	True	True	True	True					saved parameter set
Gain	Real	1.0	Non-retain	True	True	True	True					proportional gain
Ti	Real	20.0	Non-retain	True	True	True	True					reset time
Td	Real	0.0	Non-retain	True	True	True	True					derivative time
TdFiltRatio	Real	0.2	Non-retain	True	True	True	True					filter coefficient for derivative part
PWeighting	Real	1.0	Non-retain	True	True	True	True					weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	Non-retain	True	True	True	True					weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	Non-retain	True	True	True	True					PID Controller cycle time
PIDSelfTune	PID_Compact-SelfTune		Non-retain	True	True	True	True					data set for self tuning
SUT	PID_Compact_SUT		Non-retain	True	True	True	True					data set for start up tuning
CalculateParams	Bool	false	Non-retain	True	True	True	False					recalculate control parameters with parameters of startup tuning
TuneRule	Int	0	Non-retain	True	True	True	True					tuning rule for SUT (0-CHR PID,1-CHR PI)
State	Int	0	Non-retain	True	False	True	False					current phase of start up tuning
TIR	PID_Compact_TIR		Non-retain	True	True	True	True					data set for tuning in run
RunIn	Bool	false	Non-retain	True	True	True	False					activate run in setpoint without controling
CalculateParams	Bool	false	Non-retain	True	True	True	False					recalculate control parameters with parameters of tuning in run
TuneRule	Int	0	Non-retain	True	True	True	True					tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
State	Int	0	Non-retain	True	False	True	False					current phase of tuning in run
PIDCtrl	PID_Compact-Control		Non-retain	True	True	True	True					data for controling part
PIDInit	Bool	false	Non-retain	True	True	True	False					initialization of controller
IntegralSum	Real	0.0	Non-retain	True	True	True	False					signal of integral part
Retain	PID_CompactRetain		Retain	True	True	True	True					retain data
CtrlParams	PID_Compact-ControlParams		Retain	True	True	True	True					actual parameter set

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Gain	Real	1.0	Retain	True	True	True	True		proportional gain
Ti	Real	20.0	Retain	True	True	True	True		reset time
Td	Real	0.0	Retain	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	Retain	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	Retain	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	Retain	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	Retain	True	True	True	True		PID Controller cycle time

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

R_TRIG [FB1001]

R_TRIG Properties							
General							
Name	R_TRIG	**Number**	1001	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	R_TRIG				

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	Non-retain	True	True	True	False		
Output									
Q	Bool	false	Non-retain	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	Non-retain	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

R_TRIG_DB [DB4]

R_TRIG_DB Properties							
General							
Name	R_TRIG_DB	**Number**	4	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	R_TRIG				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

F_TRIG [FB1002]

F_TRIG Properties							
General							
Name	F_TRIG	**Number**	1002	**Type**	FB	**Language**	SCL
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	F_TRIG				

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	Non-retain	True	True	True	False		
Output									
Q	Bool	false	Non-retain	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	Non-retain	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

F_TRIG_DB [DB5]

F_TRIG_DB Properties							
General							
Name	F_TRIG_DB	**Number**	5	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	F_TRIG				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

R_TRIG_DB_1 [DB6]

R_TRIG_DB_1 Properties							
General							
Name	R_TRIG_DB_1	**Number**	6	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	R_TRIG				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Program blocks / System blocks / Program resources

F_TRIG_DB_1 [DB7]

F_TRIG_DB_1 Properties							
General							
Name	F_TRIG_DB_1	**Number**	7	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	BIT
Version	1.0	**User-defined ID**	F_TRIG				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
CLK	Bool	false	False	True	True	True	False		
Output									
Q	Bool	false	False	True	True	True	False		
InOut									
Static									
Stat_Bit	Bool	false	False	True	True	True	False		

PLC_2 [CPU 1211C DC/DC/DC] / Technology objects

PID_Compact_1 [DB2]

PID_Compact_1 Properties							
General							
Name	PID_Compact_1	**Number**	2	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	COMPPID
Version	2.3	**User-defined ID**	PID_Cmpt				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Setpoint	Real	0.0	False	True	True	True	False		controller setpoint input
Input	Real	0.0	False	True	True	True	False		current value from process in REAL format
Input_PER	Int	0	False	True	True	True	False		current value from peripheral input
Disturbance	Real	0.0	False	True	True	True	False		disturbance intrusion
ManualEnable	Bool	false	False	True	True	True	False		activate manual value to overwrite output value
ManualValue	Real	0.0	False	True	True	True	False		manual value
ErrorAck	Bool	false	False	True	True	True	False		reset error message
Reset	Bool	false	False	True	True	True	False		reset the controller
ModeActivate	Bool	false	False	True	True	True	False		enable mode
Output									
ScaledInput	Real	0.0	False	True	False	True	False		current value after scaling
Output	Real	0.0	False	True	False	True	False		output value in REAL format
Output_PER	Int	0	False	True	False	True	False		analog output value
Output_PWM	Bool	false	False	True	False	True	False		pulse width modulated output value
SetpointLimit_H	Bool	false	False	True	False	True	False		setpoint reached upper limit
SetpointLimit_L	Bool	false	False	True	False	True	False		setpoint reached lower limit
InputWarning_H	Bool	false	False	True	False	True	False		current value reached upper warning level
InputWarning_L	Bool	false	False	True	False	True	False		current value reached lower warning level
State	Int	0	False	True	False	True	False		current mode of operation (0-Inactive, 1-SUT, 2-TIR, 3-Automatic, 4-Manual, 5-Substitute output)
Error	Bool	false	False	True	False	True	False		error flag
ErrorBits	DWord	16#0	True	True	False	True	False		error message
InOut									
Mode	Int	4	True	True	True	True	False		mode selection
Static									
InternalDiagnostic	DWord	0	False	False	False	False	False		internal diagnostic and version handling
InternalVersion	DWord	DW#16#02030003	False	True	False	True	False		version of controller
InternalRTVersion	DWord	0	False	False	False	False	False		version of runtime
IntegralResetMode	Int	4	False	True	True	True	True		0 smooth, 1 clear, 2 keep, 3 overwrite initial output, 4 like setpoint change
OverwriteInitialOutputValue	Real	0.0	False	True	True	True	False		initialisation of output value for override control
RunModeByStartup	Bool	true	False	True	True	True	True		activate Mode after CPU restart
LoadBackUp	Bool	false	False	True	True	True	False		restore last parameter set
SetSubstituteOutput	Bool	true	False	True	True	True	True		assignment of output value in State = 5 (FALSE - last valid value, TRUE - SubstituteOutput)
PhysicalUnit	Int	0	False	True	False	True	True		unit of measurement of the process value and setpoint
PhysicalQuantity	Int	0	False	True	False	True	True		physical quantity of the process value and setpoint
ActivateRecoverMode	Bool	true	False	True	True	True	True		FALSE - go to inactive by error, TRUE - activate error treatment
Warning	DWord	16#0	True	True	False	True	False		warning message
WarningInternal	DWord	16#0	True	True	False	True	False		warning message
Progress	Real	0.0	False	True	False	True	False		progress of current phase in percent
CurrentSetpoint	Real	0.0	False	True	False	True	False		current active setpoint value
CancelTuningLevel	Real	10.0	False	True	True	True	True		cancel level for setpoint change during tuning
SubstituteOutput	Real	0.0	False	True	True	True	True		substitute output value in case of error
Config	PID_Compact-Config		False	True	True	True	True		configuration data set
InputPerOn	Bool	true	False	True	True	True	True		activate peripheral input
InvertControl	Bool	false	False	True	True	True	True		invert control direction
InputUpperLimit	Real	120.0	False	True	True	True	True		input (process value) upper limit
InputLowerLimit	Real	0.0	False	True	True	True	True		input (process value) lower limit

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
InputUpperWarning	Real	3.402822e+38	False	True	True	True	True		input (process value) upper level warning
InputLowerWarning	Real	-3.402822e+38	False	True	True	True	True		input (process value) lower level warning
OutputUpperLimit	Real	100.0	False	True	True	True	True		output value upper limit
OutputLowerLimit	Real	0.0	False	True	True	True	True		output value lower limit
SetpointUpperLimit	Real	3.402822e+38	False	True	True	True	True		setpoint upper limit value
SetpointLowerLimit	Real	-3.402822e+38	False	True	True	True	True		setpoint lower limit value
MinimumOnTime	Real	0.0	False	True	True	True	True		PWM minimum on time
MinimumOffTime	Real	0.0	False	True	True	True	True		PWM minimum off time
InputScaling	PID_Scaling		False	True	True	True	True		input scaling
UpperPointIn	Real	27648.0	False	True	True	True	True		high value (input range of scaling)
LowerPointIn	Real	0.0	False	True	True	True	True		low value (input range of scaling)
UpperPointOut	Real	100.0	False	True	True	True	True		high value (output range of scaling)
LowerPointOut	Real	0.0	False	True	True	True	True		low value (output range of scaling)
CycleTime	PID_CycleTime		False	True	True	True	True		data set for cycle time estimation
StartEstimation	Bool	true	False	True	True	True	False		start automatic estimation of call cycle time
EnEstimation	Bool	true	False	True	True	True	True		enable estimation of call cycle time
EnMonitoring	Bool	true	False	True	True	True	True		enable monitoring of call cycle time
Value	Real	0.1	False	True	True	True	True		call cycle time
CtrlParamsBackUp	PID_Compact-ControlParams		False	True	True	True	True		saved parameter set
Gain	Real	1.0	False	True	True	True	True		proportional gain
Ti	Real	20.0	False	True	True	True	True		reset time
Td	Real	0.0	False	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	False	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	False	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	False	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	False	True	True	True	True		PID Controller cycle time
PIDSelfTune	PID_Compact-SelfTune		False	True	True	True	True		data set for self tuning
SUT	PID_Compact_SUT		False	True	True	True	True		data set for start up tuning
CalculateParams	Bool	false	False	True	True	True	False		recalculate control parameters with parameters of startup tuning
TuneRule	Int	0	False	True	True	True	True		tuning rule for SUT (0-CHR PID,1-CHR PI)
State	Int	0	False	True	False	True	False		current phase of start up tuning
TIR	PID_Compact_TIR		False	True	True	True	True		data set for tuning in run
RunIn	Bool	false	False	True	True	True	False		activate run in setpoint without controling
CalculateParams	Bool	false	False	True	True	True	False		recalculate control parameters with parameters of tuning in run
TuneRule	Int	0	False	True	True	True	True		tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
State	Int	0	False	True	False	True	False		current phase of tuning in run
PIDCtrl	PID_Compact-Control		False	True	True	True	True		data for controling part
PIDInit	Bool	false	False	True	True	True	False		initialization of controller
IntegralSum	Real	0.0	False	True	True	True	False		signal of integral part
Retain	PID_CompactRetain		True	True	True	True	True		retain data
CtrlParams	PID_Compact-ControlParams		True	True	True	True	True		actual parameter set
Gain	Real	1.0	True	True	True	True	True		proportional gain
Ti	Real	20.0	True	True	True	True	True		reset time
Td	Real	0.0	True	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	True	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	True	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	True	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	True	True	True	True	True		PID Controller cycle time

PLC_2 [CPU 1211C DC/DC/DC] / Technology objects

PID_Compact_2 [DB3]

PID_Compact_2 Properties							
General							
Name	PID_Compact_2	**Number**	3	**Type**	DB	**Language**	DB
Numbering	Automatic						
Information							
Title		**Author**	SIMATIC	**Comment**		**Family**	COMPPID
Version	2.3	**User-defined ID**	PID_Cmpt				

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision				Comment
Input												
Setpoint	Real	0.0	False	True	True	True	False					controller setpoint input
Input	Real	0.0	False	True	True	True	False					current value from process in REAL format
Input_PER	Int	0	False	True	True	True	False					current value from peripheral input
Disturbance	Real	0.0	False	True	True	True	False					disturbance intrusion
ManualEnable	Bool	false	False	True	True	True	False					activate manual value to overwrite output value
ManualValue	Real	0.0	False	True	True	True	False					manual value
ErrorAck	Bool	false	False	True	True	True	False					reset error message
Reset	Bool	false	False	True	True	True	False					reset the controller
ModeActivate	Bool	false	False	True	True	True	False					enable mode
Output												
ScaledInput	Real	0.0	False	True	False	True	False					current value after scaling
Output	Real	0.0	False	True	False	True	False					output value in REAL format
Output_PER	Int	0	False	True	False	True	False					analog output value
Output_PWM	Bool	false	False	True	False	True	False					pulse width modulated output value
SetpointLimit_H	Bool	false	False	True	False	True	False					setpoint reached upper limit
SetpointLimit_L	Bool	false	False	True	False	True	False					setpoint reached lower limit
InputWarning_H	Bool	false	False	True	False	True	False					current value reached upper warning level
InputWarning_L	Bool	false	False	True	False	True	False					current value reached lower warning level
State	Int	0	False	True	False	True	False					current mode of operation (0-Inactive, 1-SUT, 2-TIR, 3-Automatic, 4-Manual, 5-Substitute output)
Error	Bool	false	False	True	False	True	False					error flag
ErrorBits	DWord	16#0	True	True	False	True	False					error message
InOut												
Mode	Int	3	True	True	True	True	False					mode selection
Static												
InternalDiagnostic	DWord	0	False	False	False	False	False					internal diagnostic and version handling
InternalVersion	DWord	DW#16#02030003	False	True	False	True	False					version of controller
InternalRTVersion	DWord	0	False	False	False	False	False					version of runtime
IntegralResetMode	Int	4	False	True	True	True	True					0 smooth, 1 clear, 2 keep, 3 overwrite initial output, 4 like setpoint change
OverwriteInitialOutputValue	Real	0.0	False	True	True	True	False					initialisation of output value for override control
RunModeByStartup	Bool	true	False	True	True	True	True					activate Mode after CPU restart
LoadBackUp	Bool	false	False	True	True	True	False					restore last parameter set
SetSubstituteOutput	Bool	true	False	True	True	True	True					assignment of output value in State = 5 (FALSE - last valid value, TRUE - SubstituteOutput)
PhysicalUnit	Int	0	False	True	False	True	True					unit of measurement of the process value and setpoint
PhysicalQuantity	Int	1	False	True	False	True	True					physical quantity of the process value and setpoint
ActivateRecoverMode	Bool	true	False	True	True	True	True					FALSE - go to inactive by error, TRUE - activate error treatment
Warning	DWord	16#0	True	True	False	True	False					warning message
WarningInternal	DWord	16#0	True	True	False	True	False					warning message
Progress	Real	0.0	False	True	False	True	False					progress of current phase in percent
CurrentSetpoint	Real	0.0	False	True	False	True	False					current active setpoint value
CancelTuningLevel	Real	10.0	False	True	True	True	True					cancel level for setpoint change during tuning
SubstituteOutput	Real	0.0	False	True	True	True	True					substitute output value in case of error
Config	PID_Compact-Config		False	True	True	True	True					configuration data set
InputPerOn	Bool	FALSE	False	True	True	True	True					activate peripheral input
InvertControl	Bool	FALSE	False	True	True	True	True					invert control direction
InputUpperLimit	Real	120.0	False	True	True	True	True					input (process value) upper limit
InputLowerLimit	Real	0.0	False	True	True	True	True					input (process value) lower limit

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
InputUpperWarning	Real	3.402822e+38	False	True	True	True	True		input (process value) upper level warning
InputLowerWarning	Real	-3.402822e+38	False	True	True	True	True		input (process value) lower level warning
OutputUpperLimit	Real	100.0	False	True	True	True	True		output value upper limit
OutputLowerLimit	Real	0.0	False	True	True	True	True		output value lower limit
SetpointUpperLimit	Real	3.402822e+38	False	True	True	True	True		setpoint upper limit value
SetpointLowerLimit	Real	-3.402822e+38	False	True	True	True	True		setpoint lower limit value
MinimumOnTime	Real	0.0	False	True	True	True	True		PWM minimum on time
MinimumOffTime	Real	0.0	False	True	True	True	True		PWM minimum off time
InputScaling	PID_Scaling		False	True	True	True	True		input scaling
UpperPointIn	Real	27648.0	False	True	True	True	True		high value (input range of scaling)
LowerPointIn	Real	0.0	False	True	True	True	True		low value (input range of scaling)
UpperPointOut	Real	100.0	False	True	True	True	True		high value (output range of scaling)
LowerPointOut	Real	0.0	False	True	True	True	True		low value (output range of scaling)
CycleTime	PID_CycleTime		False	True	True	True	True		data set for cycle time estimation
StartEstimation	Bool	true	False	True	True	True	False		start automatic estimation of call cycle time
EnEstimation	Bool	true	False	True	True	True	True		enable estimation of call cycle time
EnMonitoring	Bool	true	False	True	True	True	True		enable monitoring of call cycle time
Value	Real	0.1	False	True	True	True	True		call cycle time
CtrlParamsBackUp	PID_CompactControlParams		False	True	True	True	True		saved parameter set
Gain	Real	1.0	False	True	True	True	True		proportional gain
Ti	Real	20.0	False	True	True	True	True		reset time
Td	Real	0.0	False	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	False	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	False	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	False	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	False	True	True	True	True		PID Controller cycle time
PIDSelfTune	PID_CompactSelfTune		False	True	True	True	True		data set for self tuning
SUT	PID_Compact_SUT		False	True	True	True	True		data set for start up tuning
CalculateParams	Bool	false	False	True	True	True	False		recalculate control parameters with parameters of startup tuning
TuneRule	Int	0	False	True	True	True	True		tuning rule for SUT (0-CHR PID,1-CHR PI)
State	Int	0	False	True	False	True	False		current phase of start up tuning
TIR	PID_Compact_TIR		False	True	True	True	True		data set for tuning in run
RunIn	Bool	false	False	True	True	True	False		activate run in setpoint without controling
CalculateParams	Bool	false	False	True	True	True	False		recalculate control parameters with parameters of tuning in run
TuneRule	Int	0	False	True	True	True	True		tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
State	Int	0	False	True	False	True	False		current phase of tuning in run
PIDCtrl	PID_CompactControl		False	True	True	True	True		data for controling part
PIDInit	Bool	false	False	True	True	True	False		initialization of controller
IntegralSum	Real	0.0	False	True	True	True	False		signal of integral part
Retain	PID_CompactRetain		True	True	True	True	True		retain data
CtrlParams	PID_CompactControlParams		True	True	True	True	True		actual parameter set
Gain	Real	50.0	True	True	True	True	True		proportional gain
Ti	Real	0.0	True	True	True	True	True		reset time
Td	Real	0.0	True	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	True	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	True	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	True	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	True	True	True	True	True		PID Controller cycle time

PLC_2 [CPU 1211C DC/DC/DC] / PLC tags / Default tag table [35]

PLC tags

PLC tags

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	System_Byte	Byte	%MB1	False	True	True	True					
	FirstScan	Bool	%M1.0	False	True	True	True					
	DiagStatusUpdate	Bool	%M1.1	False	True	True	True					
	AlwaysTRUE	Bool	%M1.2	False	True	True	True					
	AlwaysFALSE	Bool	%M1.3	False	True	True	True					

PLC_2 [CPU 1211C DC/DC/DC] / PLC tags / Default tag table [35]

User constants

User constants				
	Name	Data type	Value	Comment

PLC_2 [CPU 1211C DC/DC/DC] / PLC tags / Default tag table_1 [52]

PLC tags

PLC tags

	Name	Data type	Address	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Supervision				Comment
	TERMOPARA_IN	Int	%IW64	False	True	True	True					
	POWER_ON	Bool	%M0.0	False	True	True	True					
	EMERGENCY	Bool	%M0.1	False	True	True	True					
	TEMPERATURA	Real	%MD0	False	True	True	True					
	HEATER_ON	Bool	%Q0.0	False	True	True	True					
	TEMPERATURA_SCALED	Real	%MD1	False	True	True	True					
	FAN_ON	Bool	%Q0.2	False	True	True	True					

PLC_2 [CPU 1211C DC/DC/DC] / PLC tags / Default tag table_1 [52]

User constants

User constants				
	Name	Data type	Value	Comment
	None(1)	Pip	65535	
	Automatic update(1)	Pip	0	
	PIP 1(1)	Pip	1	
	PIP 2(1)	Pip	2	
	PIP 3(1)	Pip	3	
	PIP 4(1)	Pip	4	
	PIP 5	Pip	5	
	PIP 6	Pip	6	
	PIP 7	Pip	7	
	PIP 8	Pip	8	
	PIP 9	Pip	9	
	PIP 10	Pip	10	
	PIP 11	Pip	11	
	PIP 12	Pip	12	
	PIP 13	Pip	13	
	PIP 14	Pip	14	
	PIP 15	Pip	15	
	PIP 16	Pip	16	
	PIP 17	Pip	17	
	PIP 18	Pip	18	
	PIP 19	Pip	19	
	PIP 20	Pip	20	
	PIP 21	Pip	21	
	PIP 22	Pip	22	
	PIP 23	Pip	23	
	PIP 24	Pip	24	
	PIP 25	Pip	25	
	PIP 26	Pip	26	
	PIP 27	Pip	27	
	PIP 28	Pip	28	
	PIP 29	Pip	29	
	PIP 30	Pip	30	
	PIP 31	Pip	31	
	PIP OB Servo(1)	Pip	32768	
	Local~MC(1)	Hw_SubModule	51	
	Local~Exec(1)	Hw_SubModule	52	
	Local(1)	Hw_SubModule	49	
	Local~PROFINET_interface_1(1)	Hw_Interface	64	
	Local~PROFINET_interface_1~Port_3	Hw_Interface	67	
	Local~PROFINET_interface_1~Port_1(1)	Hw_Interface	65	
	Local~PROFINET_interface_1~Port_2	Hw_Interface	66	
	OB_Main(1)	OB_PCYCLE	1	
	Local~Common(1)	Hw_SubModule	50	
	Local~Device(1)	Hw_Device	32	
	Local~Configuration(1)	Hw_SubModule	33	

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CompactConfig

PID_CompactConfig Properties							
General							
Name	PID_CompactConfig	**Number**	1134	**Type**	UDT	**Language**	
Numbering							
Information							
Title	configuration data set	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
InputPerOn	Bool	true	True	True	True	True		activate peripheral input
InvertControl	Bool	false	True	True	True	True		invert control direction
InputUpperLimit	Real	120.0	True	True	True	True		input (process value) upper limit
InputLowerLimit	Real	0.0	True	True	True	True		input (process value) lower limit
InputUpperWarning	Real	3.402822e+38	True	True	True	True		input (process value) upper level warning
InputLowerWarning	Real	-3.402822e+38	True	True	True	True		input (process value) lower level warning
OutputUpperLimit	Real	100.0	True	True	True	True		output value upper limit
OutputLowerLimit	Real	0.0	True	True	True	True		output value lower limit
SetpointUpperLimit	Real	3.402822e+38	True	True	True	True		setpoint upper limit value
SetpointLowerLimit	Real	-3.402822e+38	True	True	True	True		setpoint lower limit value
MinimumOnTime	Real	0.0	True	True	True	True		PWM minimum on time
MinimumOffTime	Real	0.0	True	True	True	True		PWM minimum off time
▼ InputScaling	PID_Scaling		True	True	True	True		input scaling
UpperPointIn	Real	27648.0	True	True	True	True		high value (input range of scaling)
LowerPointIn	Real	0.0	True	True	True	True		low value (input range of scaling)
UpperPointOut	Real	100.0	True	True	True	True		high value (output range of scaling)
LowerPointOut	Real	0.0	True	True	True	True		low value (output range of scaling)

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_Scaling

PID_Scaling Properties							
General							
Name	PID_Scaling	**Number**	1135	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data for scaling	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
UpperPointIn	Real	27648.0	True	True	True	True		high value (input range of scaling)
LowerPointIn	Real	0.0	True	True	True	True		low value (input range of scaling)
UpperPointOut	Real	100.0	True	True	True	True		high value (output range of scaling)
LowerPointOut	Real	0.0	True	True	True	True		low value (output range of scaling)

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CycleTime

PID_CycleTime Properties							
General							
Name	PID_CycleTime	**Number**	1137	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data set for cycle time estimation	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
StartEstimation	Bool	true	True	True	True	False		start automatic estimation of call cycle time
EnEstimation	Bool	true	True	True	True	True		enable estimation of call cycle time
EnMonitoring	Bool	true	True	True	True	True		enable monitoring of call cycle time
Value	Real	0.1	True	True	True	True		call cycle time

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CompactControlParams

PID_CompactControlParams Properties							
General							
Name	PID_CompactControlParams	**Number**	1138	**Type**	UDT	**Language**	
Numbering							
Information							
Title	controling parameter set	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Gain	Real	1.0	True	True	True	True		proportional gain
Ti	Real	20.0	True	True	True	True		reset time
Td	Real	0.0	True	True	True	True		derivative time
TdFiltRatio	Real	0.2	True	True	True	True		filter coefficient for derivative part
PWeighting	Real	1.0	True	True	True	True		weigthing of proportional part in direct, feedback path
DWeighting	Real	1.0	True	True	True	True		weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	True	True	True	True		PID Controller cycle time

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CompactSelfTune

PID_CompactSelfTune Properties							
General							
Name	PID_CompactSelfTune	**Number**	1139	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data set for self tuning	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
SUT	PID_Compact_SUT		True	True	True	True		data set for start up tuning
CalculateParams	Bool	false	True	True	True	False		recalculate control parameters with parameters of startup tuning
TuneRule	Int	0	True	True	True	True		tuning rule for SUT (0-CHR PID,1-CHR PI)
State	Int	0	True	False	True	False		current phase of start up tuning
TIR	PID_Compact_TIR		True	True	True	True		data set for tuning in run
RunIn	Bool	false	True	True	True	False		activate run in setpoint without controling
CalculateParams	Bool	false	True	True	True	False		recalculate control parameters with parameters of tuning in run
TuneRule	Int	0	True	True	True	True		tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
State	Int	0	True	False	True	False		current phase of tuning in run

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_GradientEstimation

PID_GradientEstimation Properties							
General							
Name	PID_GradientEstimation	**Number**	1508	**Type**	UDT	**Language**	
Numbering							
Information							
Title	structure for gradient estimation	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_GradientParams

PID_GradientParams Properties							
General							
Name	PID_GradientParams	**Number**	1511	**Type**	UDT	**Language**	
Numbering							
Information							
Title	dataset of parameters for gradient estimation	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_StandardDeviation

PID_StandardDeviation Properties							
General							
Name	PID_StandardDeviation	**Number**	1509	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data for estimation of deviance	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_Compact_SUT

PID_Compact_SUT Properties							
General							
Name	PID_Compact_SUT	**Number**	1142	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data set for start up tuning	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
CalculateParams	Bool	false	True	True	True	False		recalculate control parameters with parameters of startup tuning
TuneRule	Int	0	True	True	True	True		tuning rule for SUT (0-CHR PID,1-CHR PI)
State	Int	0	True	False	True	False		current phase of start up tuning

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_Compact_TIR

PID_Compact_TIR Properties							
General							
Name	PID_Compact_TIR	**Number**	1143	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data set for tuning in run	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
RunIn	Bool	false	True	True	True	False		activate run in setpoint without controling
CalculateParams	Bool	false	True	True	True	False		recalculate control parameters with parameters of tuning in run
TuneRule	Int	0	True	True	True	True		tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
State	Int	0	True	False	True	False		current phase of tuning in run

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CompactControl

PID_CompactControl Properties							
General							
Name	PID_CompactControl	**Number**	1144	**Type**	UDT	**Language**	
Numbering							
Information							
Title	data for controling part	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
PIDInit	Bool	false	True	True	True	False		initialization of controller
IntegralSum	Real	0.0	True	True	True	False		signal of integral part

PLC_2 [CPU 1211C DC/DC/DC] / PLC data types / System data types

PID_CompactRetain

PID_CompactRetain Properties							
General							
Name	PID_CompactRetain	**Number**	1145	**Type**	UDT	**Language**	
Numbering							
Information							
Title	retain data	**Author**		**Comment**		**Family**	
Version		**User-defined ID**					

Name	Data type	Default value	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision				Comment
CtrlParams	PID_CompactControl-Params		True	True	True	True					actual parameter set
Gain	Real	1.0	True	True	True	True					proportional gain
Ti	Real	20.0	True	True	True	True					reset time
Td	Real	0.0	True	True	True	True					derivative time
TdFiltRatio	Real	0.2	True	True	True	True					filter coefficient for derivative part
PWeighting	Real	1.0	True	True	True	True					weigthing of proportional part in direct, feed-back path
DWeighting	Real	1.0	True	True	True	True					weigthing of derivative part in direct, feedback path
Cycle	Real	1.0	True	True	True	True					PID Controller cycle time

PLC_2 [CPU 1211C DC/DC/DC] / Watch and force tables

Force table

Name	Address	Display format	Force value	Comment
"Temperatura_DB".setpoint	%DB1.DBD14	Floating-point number	30.0	

PLC_2 [CPU 1211C DC/DC/DC] / Watch and force tables

Watch table_1

Name	Address	Display format	Modify value	Comment
"Temperatura_DB".setpoint	%DB1.DBD14	Floating-point number		
"EMERGENCY"	%M0.1	Bool		
"Temperatura_DB".ON_OFF	%DB1.DBX8.0	Bool	TRUE	
"Temperatura_DB".temperatura	%DB1.DBD0	Floating-point number		
"Temperatura_DB".cyclicMode	%DB1.DBX22.0	Bool		
"Temperatura_DB".ON_OFF_activate	%DB1.DBX22.1	Bool		

PLC_2 [CPU 1211C DC/DC/DC]

Traces

Name

PLC_2 [CPU 1211C DC/DC/DC] / Traces

Measurements

This folder is empty.

PLC_2 [CPU 1211C DC/DC/DC] / Traces

Combined measurements

Name

PLC_2 [CPU 1211C DC/DC/DC] / OPC UA communication

Server interfaces

This folder is empty.

PLC_2 [CPU 1211C DC/DC/DC]

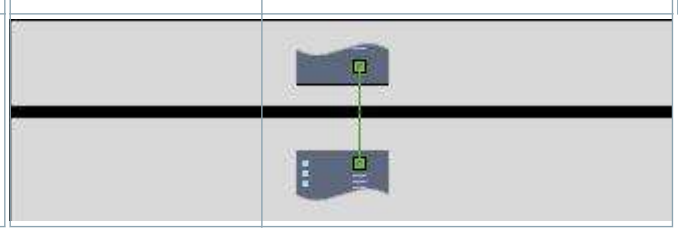
PLC alarm text lists

This folder is empty.

PLC_2 [CPU 1211C DC/DC/DC] / Local modules

PLC_2 [CPU 1211C DC/DC/DC]

PLC_2					
General\Project information					
Name	PLC_2	**Author**	praktykant	**Comment**	
Slot	1	**Rack**	0		
General\Catalog information					
Short designation	CPU 1211C DC/DC/DC	**Description**	Work memory 75 KB; 24VDC power supply with DI6 x 24VDC SINK/SOURCE, DQ4 x 24VDC and AI2 on board; 3 high-speed counters (expandable with digital signal board) and 4 pulse outputs on board; signal board expands on-board I/O; up to 3 communication modules for serial communication; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA	**Article number**	6ES7 211-1AE40-0XB0
Firmware version	V4.6		False		
General\Identification & Maintenance					
Plant designation		**Location identifier**		**Installation date**	2025-07-14 09:19:44.558
Additional information					
General\Checksums					
Text lists	FA 70 E8 75 1D 5A 8E 29	**Software**	EE 93 4A 0E 3B 14 21 07		
PROFINET interface [X1]\General					
Name	PROFINET interface_1	**Author**	praktykant	**Comment**	
PROFINET interface [X1]\General\Project information					
Name	DI 6/DQ 4_1	**Comment**		**Name**	AI 2_1
Comment					
PROFINET interface [X1]\Ethernet addresses\Interface networked with					
Subnet:	PN/IE_1				
PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)					
IP configuration	Set IP address in the project	**IP address:**	10.10.1.50	**Subnet mask:**	255.255.255.0
Use router	False				
PROFINET interface [X1]\Ethernet addresses\PROFINET					
PROFINET device name is set directly at the device	False	**Generate PROFINET device name automatically**	True	**PROFINET device name:**	plc_2
Converted name:	plcxb2d1ad	**Device number:**	0		
PROFINET interface [X1]\Time synchronization					
Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses	**Server 1**	0.0.0.0
Server 2	0.0.0.0	**Server 3**	0.0.0.0	**Server 4**	0.0.0.0
Update interval	10sec			**CPU synchronizes the modules of the device.**	No synchronization
PROFINET interface [X1]\Digital inputs\Channel0					
Channel address	I0.0	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel0					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49152	**Event name:**	0
Hardware interrupt:	0	**Rising edge0**	Rising edge0		
PROFINET interface [X1]\Digital inputs\Channel0					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49280	**Event name:**	0
Hardware interrupt:	0	**Falling edge0**	Falling edge0		
PROFINET interface [X1]\Digital inputs\Channel1					
Channel address	I0.1	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel1					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49153	**Event name:**	0
Hardware interrupt:	0	**Rising edge1**	Rising edge1		
PROFINET interface [X1]\Digital inputs\Channel1					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49281	**Event name:**	0
Hardware interrupt:	0	**Falling edge1**	Falling edge1		
PROFINET interface [X1]\Digital inputs\Channel2					
Channel address	I0.2	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel2					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49154	**Event name:**	0
Hardware interrupt:	0	**Rising edge2**	Rising edge2		
PROFINET interface [X1]\Digital inputs\Channel2					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49282	**Event name:**	0
Hardware interrupt:	0	**Falling edge2**	Falling edge2		
PROFINET interface [X1]\Digital inputs\Channel3					
Channel address	I0.3	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel3					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49155	**Event name:**	0
Hardware interrupt:	0	**Rising edge3**	Rising edge3		

PROFINET interface [X1]\Digital inputs\Channel3					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49283	**Event name:**	0
Hardware interrupt:	0	**Falling edge3**	Falling edge3		
PROFINET interface [X1]\Digital inputs\Channel4					
Channel address	I0.4	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel4					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49156	**Event name:**	0
Hardware interrupt:	0	**Rising edge4**	Rising edge4		
PROFINET interface [X1]\Digital inputs\Channel4					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49284	**Event name:**	0
Hardware interrupt:	0	**Falling edge4**	Falling edge4		
PROFINET interface [X1]\Digital inputs\Channel5					
Channel address	I0.5	**Input filters**	6.4 millisec	**Enable pulse catch**	0
PROFINET interface [X1]\Digital inputs\Channel5					
Enable rising edge detection	0	**Prefix Event Rising Edge**	49157	**Event name:**	0
Hardware interrupt:	0	**Rising edge5**	Rising edge5		
PROFINET interface [X1]\Digital inputs\Channel5					
Enable falling edge detection	0	**Prefix Event Falling Edge**	49285	**Event name:**	0
Hardware interrupt:	0	**Falling edge5**	Falling edge5		
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	**Measurement type**	Voltage	**Voltage range**	0..10 V
Smoothing	Weak (4 cycles)			**Enable overflow diagnostics**	1
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				
PROFINET interface [X1]\Digital outputs\Channel0					
Channel address	Q0.0	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel1					
Channel address	Q0.1	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel2					
Channel address	Q0.2	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Digital outputs\Channel3					
Channel address	Q0.3	**Substitute a value of 1 on a change from RUN to STOP.**	0		
PROFINET interface [X1]\Operating mode					
IO controller	True	**IO system**		**Device number**	0
IO device	False				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	0.0	**End address**	0.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	64	**End address**	67	**Organization block**	0
Process image	0				
PROFINET interface [X1]\I/O addresses\Output addresses					
Start address	0.0	**End address**	0.7	**Organization block**	0
Process image	0				
PROFINET interface [X1]\Advanced options\Interface options					
Support device replacement without exchangeable medium	True	**Permit overwriting of device names of all assigned IO devices**	False	**Use IEC V2.2 LLDP mode**	False
Keep-Alive connection monitoring:	30s				
PROFINET interface [X1]\Advanced options\Real time settings\IO communication					
Send clock:	1.000ms				
PROFINET interface [X1]\Advanced options\Real time settings\Real time options					
Calculated bandwidth for cyclic IO data:	0.000ms	**Calculated bandwidth for cyclic IO data:**	0.000%		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General					
Name	Port_1	**Author**	praktykant	**Comment**	
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:					
Local port:	PLC_2\PROFINET interface_1 [X1]\Port_1 [X1 P1]	**Medium:**	Copper	**Cable name:**	---
					

PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:					
	Monitoring of partner port is not possible	**Partner port:**	Switch_1\SCALANCE interface_1 [X1]\Port_3 [X1 P3]	**Medium:**	Copper
Cable length:					
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate					
Activate this port for use	True				
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection					
Transmission rate / duplex:	Automatic	**Monitor**	False	**Enable autonegotiation**	True
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries					
End of detection of accessible devices	False	**End of topology discovery**	False	**End of the sync domain**	False
PROFINET interface [X1]\Web server access					
Enable Web server for the IP address of this interface	False	**The Web server must also be activated in the properties of the PLC.**			
High speed counters (HSC)\HSC1\General\Enable					
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
Enable this high speed counter	0	**Enable this high speed counter**	0	**Enable this high speed counter**	0
High speed counters (HSC)\HSC1\General\Project information					
Name	HSC_1	**Comment**		**Name**	HSC_2
Comment		**Name**	HSC_3	**Comment**	
Name	HSC_4	**Comment**		**Name**	HSC_5
Comment		**Name**	HSC_6	**Comment**	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses					
Start address	1000.0	**End address**	1003.7	**Start address**	1004.0
End address	1007.7	**Organization block**	0	**Start address**	1008.0
End address	1011.7	**Organization block**	0	**Process image**	0
Start address	1012.0	**End address**	1015.7	**Organization block**	0
Process image	0	**Start address**	1016.0	**End address**	1019.7
Organization block	0	**Process image**	0	**Start address**	1020.0
End address	1023.7	**Organization block**	0	**Process image**	0
Organization block	0	**Process image**	0	**Process image**	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable					
Enable this pulse generator	0	**Enable this pulse generator**	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information					
Name	Pulse_1	**Comment**		**Name**	Pulse_2
Comment					
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses					
Start address	1000.0	**End address**	1001.7	**Start address**	1002.0
End address	1003.7	**Organization block**	0	**Organization block**	0
Process image	0	**Process image**	0		
Startup					
Startup after POWER ON	Warm restart - mode before POWER OFF	**Comparison preset to actual configuration**	Startup CPU even if mismatch	**Configuration time**	60000ms
OBs should be interruptible	1				
Cycle					
Cycle monitoring time [ms]	150ms			**Enable minimum cycle time for cyclic OBs**	0
Minimum cycle time	1ms				
Communication load					
Cycle load due to communication [%]	20%				
System and clock memory\System memory bits					
Enable the use of system memory byte	1	**Address of system memory byte (MBx)**	1	**First cycle**	%M1.0 (FirstScan)
Diagnostic status changed	%M1.1 (DiagStatusUpdate)	**Always 1 (high)**	%M1.2 (AlwaysTRUE)	**Always 0 (low)**	%M1.3 (AlwaysFALSE)
System and clock memory\Clock memory bits					
Enable the use of clock memory byte	0	**Address of clock memory byte (MBx)**	0	**10 Hz clock**	
5 Hz clock		**2.5 Hz clock**		**2 Hz clock**	
1.25 Hz clock		**1 Hz clock**		**0.625 Hz clock**	
0.5 Hz clock					
Web server\General					
Activate Web server on all modules of this device	False	**Permit access only with HTTPS**	True		
Web server\Automatic update					
Enable automatic update	True	**Update interval**	0s		

Web server\User management

User name	User rights
Everybody	

Web server\User-defined web pages

Application name	HTML source path	Default HTML page	Files with dynamic content	Web DB number	Fragment DB number
		index.htm	.htm;.html	333	334

Web server\Overview of interfaces

Device	Interface	Enabled web server access
PLC_2	PROFINET interface_1	False

User interface languages	
Assign project language	**User interface languages**
English (United States)	German
English (United States)	English
English (United States)	French
English (United States)	Spanish
English (United States)	Italian
English (United States)	Chinese (simplified)

Time of day\Local time					
Time zone	(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna				
Time of day\Daylight saving time					
Activate daylight saving time	1	**Difference between standard and daylight saving time**	60min		
Time of day\Daylight saving time\Start of daylight saving time					
First week of the month:	Last		Sunday	**in**	March
at	1:00 a.m.				
Time of day\Daylight saving time\Start of standard time					
	Last		Sunday	**in**	October
at	2:00 a.m.				
Protection & Security					
Level of protection	No protection				
Protection & Security\Connection mechanisms					
Permit access with PUT/GET communication from remote partner	True				
Protection & Security\Security event					
Summarize diagnostics in case of high message volume	True	**Length of an interval**	20	**Unit**	seconds
Protection & Security\External load memory					
Disable copying from internal load memory to external load memory	False				
Advanced configuration\DnsParameterConfigurationMenu					
The Tree-Node DnsConfigurationMenu was not filled by some ACF					
Advanced configuration\Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).					
Activate SNMP	False				
Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				

Connection resources

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_2 [CPU 1211C DC/DC/DC] - Configured
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	2	0	2
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		2	0	2
Available resources:		32	34	66

Overview of addresses\Overview of addresses\Overview of addresses					
Inputs	True	**Outputs**	True	**Address gaps**	False
Slot	True				

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	0	DI 6/DQ 4_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	1 Bytes	-	0	1 1
O	0	0	DI 6/DQ 4_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	1 Bytes	-	0	1 1
I	64	67	AI 2_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_2 [CPU 1211C DC/DC/DC]	-	2 Bytes	-	0	1 35